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Chapter 2: natural # as an ordered set

Def 1 & Lemma 1

trichotomy law

$x, y \in S$ only 1 holds: $x < y$, $x = y$, $y < x$

weak

$x, y \in S$ at least 1 holds: $x < y$, $x = y$, $y < x$

The weak trichotomy law implies the trichotomy law.

Ex 2

Claim: If $(S, <)$ satisfies 1., 2., 3. of Lemma 1 then it is a strict linear order.

↳ It satisfies 1 & 2 of Def. 1

1. of Lemma $\Rightarrow$ 1. of Def.

Let $x, y \in S$. Suppose at least 1 of $x < y$, $x = y$, $y < x$ holds. We are worried that two of these might hold.

6:52 PM Thu Feb 5 5G 20%

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1. of Lemma $\Rightarrow$ 1. of Def.

Let $x, y \in S$. Suppose at least 1 of $x < y$, $x = y$, $y < x$ holds. We are worried that two of these might hold.

① Suppose $x < y$ & $x = y$. $(\Leftrightarrow y = x)$

This implies that $x < y = x$. Thus $x < x$.

By the anti-reflexive law this is a contradiction.

* Other cases are hw, done similarly *

Def. 2

Associated to a strict linear order $<$ on a set S, we define a nonstrict order relation $\leq$ by defining for all $x, y \in S$:

$$x \leq y \overset{\text{def}}{\Longleftrightarrow} x < y \text{ or } x = y$$

Thm. 2

6:52 PM Thu Feb 5 5G 20%

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Thm. 2

If $x < y$ & $y \leq z$ then $x < z$. Also, if $x \leq y$ & $y < z$ then $x < z$.

mixed transitivity

Pf.

Suppose $x < y$ & $y \leq z$. We have two cases. ($y = z$, $y < z$)

If $y < z$, then transitivity shows $x < z$. ✓

If $y = z$, then $x < y = z$ so $x < z$. ✓

$\square$